

Fundamentals of Data Science for Research

Data Science Lifecycle | Ethics | Tidy Data | Measurement | Reproducibility

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What is Data Science?

- **Interdisciplinary field** combining statistics, computing, and domain knowledge
- ✦ Turns data into **insights and solutions**
- 🌐 Used in **research, science, business, and policy**

Note: Link the concept to evidence-based research.

Data Science Lifecycle



Note: Introduce lifecycle as research workflow.

Stage 1: Problem Definition

🚩 Define **research goals**

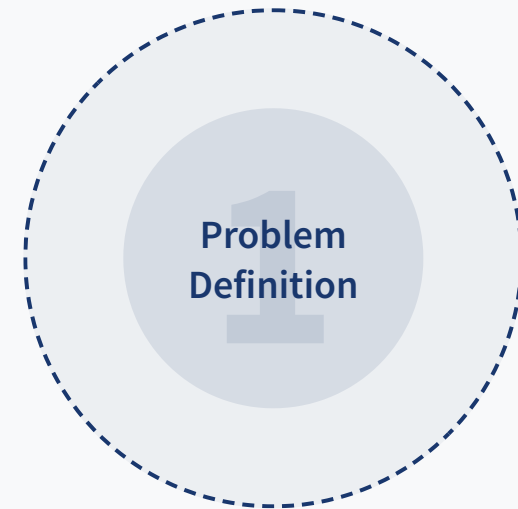
🔍 Identify **variables**

📖 Review **literature**

✅ Ensure **feasibility**

💡 Example

Predict diabetes risk using demographic + lifestyle data



Stage 2: Data Collection

🔬 **Primary** (surveys, experiments)

☁️ **Secondary** (datasets, APIs)

✂️ **Sampling methods**

✅ **Data quality checks**

💡 Example

Student performance dataset from school records



Stage 3: Data Cleaning & Preprocessing

-  Handle missing values
-  Detect outliers
-  Convert formats & fix data types
-  Feature engineering

Example

Replace missing "age" values using median



Stage 4: Exploratory Data Analysis (EDA)

- Summary statistics
- Visualizations
- Pattern detection
- Correlation analysis

Example

Scatterplot of BMI vs. glucose

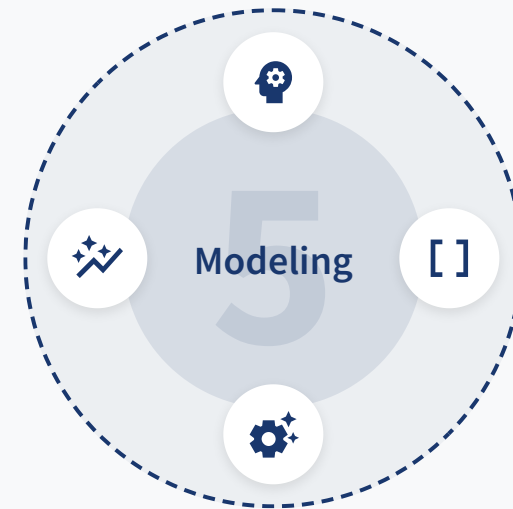


Stage 5: Modeling

- 🔗 Choose appropriate model
- ↕ Train/Validation/Test split
- ⚙️ Hyperparameter tuning
- ↔️ Statistical vs. ML models

💡 Example

Logistic Regression for classification



Stage 6: Evaluation

▲ Classification metrics: Accuracy, Precision, Recall, F1

↗ Regression metrics: MAE, RMSE, R^2

📊 Importance of confusion matrix

💡 Example

High accuracy, low recall → missing positive cases

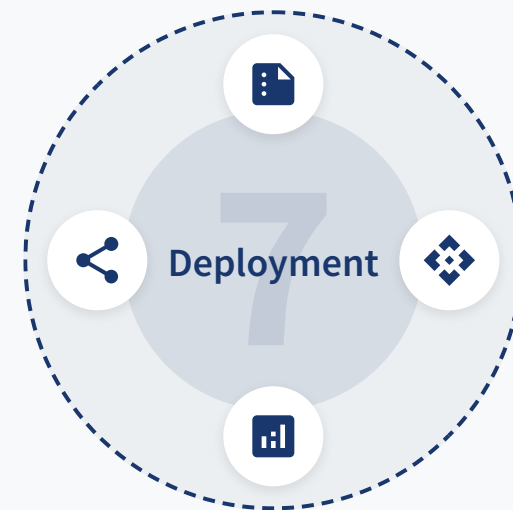


Stage 7: Deployment & Communication

-  Report findings
-  Publish research
-  Deploy models (API, app)
-  Create dashboards

Note

Focus on impact and decision-making



Research Fields & Use-Cases



STEM

Bioinformatics, engineering, environmental science



Social Sciences

Sentiment analysis, survey analysis, social networks



Business

Market analysis, customer segmentation, financial modeling



Health & Medicine

Disease risk modeling, medical imaging, epidemiology



Show diversity of Data Science research

STEM Use-Cases



Bioinformatics

Gene expression analysis

RNA-seq

Protein structure

Drug discovery



Engineering

Predictive maintenance

Sensor data

Fault detection

Optimization



Environmental Science

Air quality prediction

Climate modeling

Pollution tracking

Remote sensing

Example

 Using sensors to detect machine failure

Social Science Use-Cases



Sentiment Analysis

Extracting opinions and emotions from text data

Social media

Product reviews

Public opinion



Survey Analysis

Extracting insights from structured questionnaires

Market research

Opinion polls

Factor analysis



Social Network Modeling

Analyzing relationships and influence patterns

Community detection

Information flow

Influence modeling

Example

 Effect of SES on student performance

Health & Medical Applications



Disease Risk Modeling

Predicting health outcomes using patient data

Risk factors

Clinical decision support

Personalized medicine



Medical Image Classification

Analyzing medical images for diagnosis

Radiology

Pathology

Deep learning



Epidemiological Forecasting

Predicting disease spread patterns

Time series analysis

Spatial modeling

Public health

Example



COVID-19 spread modeling using time series

Characteristics of a Good Research Question



Specific & Measurable

Clear, focused, and quantifiable parameters



Novel but Feasible

Original yet achievable with available resources



Data-Driven

Grounded in evidence and empirical observation



Ethical & Relevant

Respects participants and addresses important issues

Example

 Does daily walking lower BP among adults 30–60?

PICO & FINER Frameworks

PICO

- P** **Population**
Who is being studied?
- I** **Intervention**
What is being tested?
- C** **Comparison**
What is the alternative?
- O** **Outcome**
What is being measured?

FINER

- F** **Feasible**
Is it practical to conduct?
- I** **Interesting**
Will it attract interest?
- N** **Novel**
Does it add to existing knowledge?
- E** **Ethical**
Does it follow ethical guidelines?
- R** **Relevant**
Does it address important issues?

Example

 Compare exercise vs. no exercise on blood pressure

Types of Variables



Independent

The variable that is **manipulated** or changed to observe its effect



Dependent

The variable that is **measured** to see the effect of the independent variable



Control

Variables that are **kept constant** to isolate the effect of the independent variable



Confounders

Variables that influence both the independent and dependent variables, potentially **distorting** their relationship

Example



Study hours (IV), Exam score (DV), IQ (confounder)

Measurement Scales



Nominal Categories

Data with **no order** or ranking

Examples: gender, color, country of origin



Ordinal Ordered

Data with **meaningful order** but unequal intervals

Examples: Likert scale, education level, customer satisfaction



Interval Equal intervals

Ordered data with **equal intervals** but no true zero

Examples: Celsius temperature, IQ scores, calendar years



Ratio True zero

Ordered data with **equal intervals** and a meaningful zero

Examples: height, weight, income, age

Key Point



Scale determines appropriate statistical tests

Tidy Data Principles

||| One variable per column

— One observation per row

☐ One value per cell

💡 Example

↕ Converting wide format to long format

☐ Tidy Data Structure

Name	Treatment	Result
John Smith	a	-
Jane Doe	a	16
Mary Johnson	a	3
John Smith	b	2
Jane Doe	b	11

Tidy datasets are all alike; every messy dataset is messy in its own way.

— Hadley Wickham

Data Ethics & Bias



Privacy & Consent

Respecting **individual rights** and obtaining proper permission for data use



Data Anonymity

Removing **personal identifiers** to protect subject identity



Fairness & Equity

Ensuring **equal treatment** and avoiding discriminatory outcomes



Biases

Sampling, measurement, and **algorithmic** biases that can distort results

Example

 Face recognition bias on darker skin tones

Reproducibility in Research



Version Control

Track changes with **Git** to maintain code history



Notebooks

Use **Jupyter/RMarkdown** for documented analysis



Random Seed Control

Set **consistent seeds** for reproducible randomization



Documentation

Create **readme files** and clear documentation

Example

 Using GitHub + Zenodo for dataset DOI

Dataset Sanity Checks



Valid Ranges

Verify values fall within **expected boundaries**



Missing Values

Identify and **handle nulls** appropriately



Duplicate Entries

Detect and **remove redundant records**



Data Type Consistency

Ensure **uniform formatting** across the dataset



Domain Alignment

Validate that data **matches domain knowledge** and expected patterns

Example



Remove records with age = 150 (invalid)

Summary & Takeaways



Lifecycle guides the research workflow



Good questions → good data → good insights



Ethics and **reproducibility** ensure trustworthy research



Always **clean**, **validate**, and **document** data

♥ Thank you!